

RAGULAN KARUNANITHI

Mechanical Engineering Graduate

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EXPERIENCE

MSc Student

Mopko National University

March 15(2026)Onwards Mokpo,Korea

- CFD analysis of a two-stage pump as turbine evaluates hydraulic performance and internal flow characteristics. The expected efficiency is approximately 72, with higher flow instability under off-design conditions and improved energy distribution near the best efficiency point.

Instructor

SLIIT-Malabe

May 15(2025)- Feb 15(2026) Malabe,Colombo

- Instruct laboratory sessions, assess student reports, participate in departmental meetings, and support academic responsibilities and administrative tasks as required.

Research Assistant

Joint Research between University Of Peradeniya and Mopko National University

Sep 15(2024) - May 15(2025) Malabe,Colombo

- This joint research focuses on creating a digital twin for centrifugal pumps, starting with 3D modeling, ICEM meshing and ANSYS CFX simulations, then advancing to reduced-order modeling and applications.

Machine Learning Intern

International distillers Ltd (IDL)

July 2023 - Oct 2023 Malabe,Colombo

- Developed hardware system with ESP8266 for bottle counting, stored data in MySQL, and created a Python Tkinter GUI app for universal data analysis in manufacturing .
- Electrical engineer designing motor starting panels & conveyor systems, ensuring safety, efficiency, and compliance with industry standards.

Mechanical Engineering Intern on MEP

National Water Supply

July 2022 - Sep 2022 Kandy, Sri Lanka

- worked on developing and optimizing water treatment processes.
- conducted pump tests for centrifugal pump.
- Worked on wastewater treatment plant operations .
- Designed a centrifugal pump and analyzed input-output flow conditions using ANSYS CFX for performance optimization.

MY LIFE PHILOSOPHY

"If you don't have any shadows, you're not standing in the light."

SUMMARY

An adaptable and responsible graduate seeking an entry-level position in engineering that will utilize the technical and creative skills developed as an undergraduate during professional training sessions. During my degree, I successfully combined my studies with work and other commitments showing myself to be self-motivated, organized, and capable of working under pressure. I have a clear, logical mind with a practical approach to problem-solving and a drive to see things through to completion. I enjoy working on my own initiative or in a team. I am a reliable, trustworthy, hardworking individual with a genuine interest in engineering .

MOST PROUD OF



Won Professor T.D.M.A.Samuel Prize for Enthusiasm and Excellence in Engineering Mathematics among 415 Engineering undergraduates



Winning grade 11 province-level Olympiad

STRENGTHS

Hard-working(18/24)

Team Building Leadership

Problem solving

Statistical analysis

TECHNICAL SKILLS

Solidworks



Python



Matlab



PROJECTS

Optimizing biomass combustion and mitigating potassium induced slag formation with deep learning (Free-lancing project)

This project involved using CNNs for potassium content analysis and RNN-LSTM models for real-time control, achieving 98.74 accuracy and improving combustion efficiency, stability, and slag prevention in biomass power plants.

Enhancing Predictive Capabilities of Multiphysics Modelling Through Machine Learning (Final year project)

This research merges machine learning with multiphysics modeling to enhance predictive capabilities, starting with optimizing solar panel performance using available data. Through expertise in regression algorithms, ANSYS integration, and solar cell studies, the goal is to extend the model's applicability to industries facing data scarcity for predictive analysis .

Implement a hardware system to store the bottle count in a specific database and making a app for calculate the bottle count and downtime by using that database (Internship project)

The project involves hardware with ESP8266 for bottle counting, MySQL database for storage, and a Python Tkinter GUI app for universal data visualization and analysis in manufacturing plants

Designing of a Treadmill Dashboard Menu using python (First-year project-Computing)

Program a dashboard to view useful insights into a treadmill user based on the conditions at which the treadmill is operated

Designing of an Automatic Kottu Roti-making machine (Second year project using solidworks)

We implemented as a semi-autoamated Kottu Roti making machine. This product is inspired by some existing products, such as the koththu rotti cutting machine and some fried rice making machines which are used in China and japan.

Design and Development of smart Fish Box Cooler (Third year project)

For Maintaining Fish Freshness during Transportation Design and develop a fish box cooler that meets the specific needs of fishermen and is durable, easy to transport, and can maintain the temperature of the catch .

Designing a personal website (First-year project -Computing)

On this website, you will find information about me, including my background, interests, and aspirations in the field of computing.
<https://sites.google.com/d/17jfzVvy718CYc1d12YvSeT36uK3N9XPm/p/1D1e1zjcFXUGMhrekuffpGa0vmwuga74ve/edit>

Deep Learning	
Machine Learning	
NLP	
Ansyes CFX	
Ansyes Fluent	
ICEM CFD	

LANGUAGES

English	
Sinhalam	
Tamil	

EDUCATION

BSc.Eng (Mechanical Engineering)

University of Peradeniya

2019 to 2024 Kandy,Srilanka

Current GPA:3.35/4.0

ML, AI and Deep Learning

Pantech University

2023 to 2024 May Online

Coursework: OOP in python, Machine Learning projects , Deep learning Projects,AI projects and Image processing

Secondary Education

Hartley college

Jan 2009 - Aug 2017 Jaffna,Srilanka

OL - 9A's -Commerce, ICT, Tamil literature
AL- 2A,1B -Physical Science

Primary Education

Gnannasariyar college

Jan 2004 - Dec2008 Jaffna,Srilanka

REFEREES

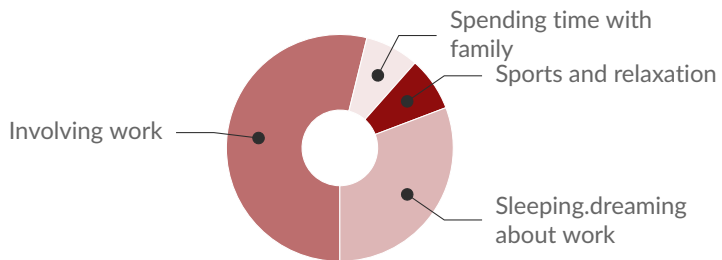
Dr.Sanjeewa Maithripala

@ Senior Lecturer, Department of Mechanical Engineering, Faculty of Engineering, University of Peradeniya, Sri Lanka

Design of a virtual pandal using Arduino using Tinkercad Simulation Tool(First-year project -Measurement)

I showcased my measurement skills, Arduino programming proficiency, and creativity in designing a virtual pandal using the Tinkercad Simulation Tool. <https://youtu.be/TrG0y8NM21M>

A DAY OF MY LIFE



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☎ +94812393600

Relationship: Lecturer, FYP Guider

Prof. Asanga Ratnaweera

@ Professor, Department of Mechanical Engineering, Faculty of Engineering, University of Peradeniya, Sri Lanka

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☎ +94773178648

Relationship: Lecturer, FYP Evaluator